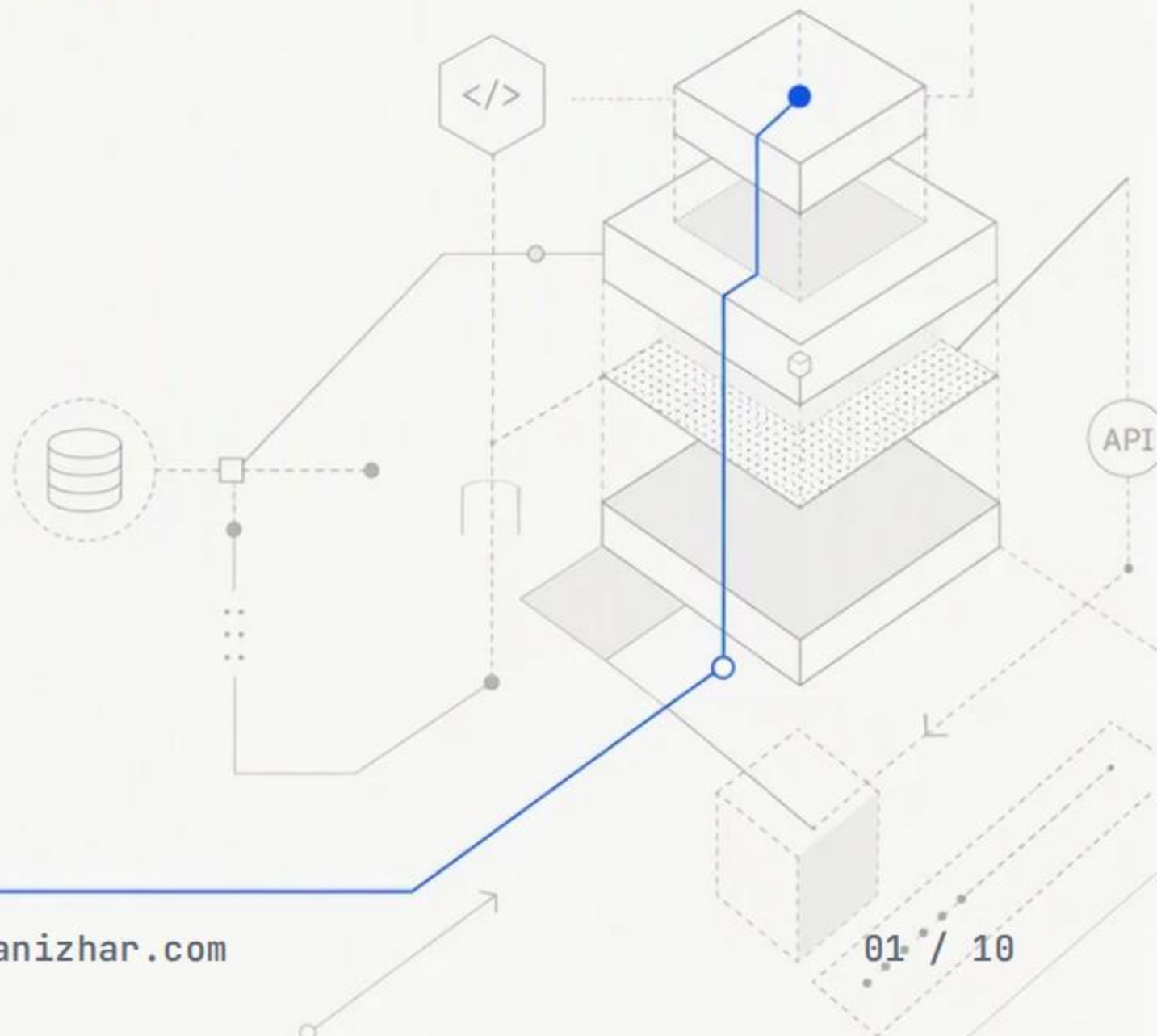


# Retrieval fails before the model does.

Five checks I run before touching a prompt  
or swapping a model.

SWIPE →

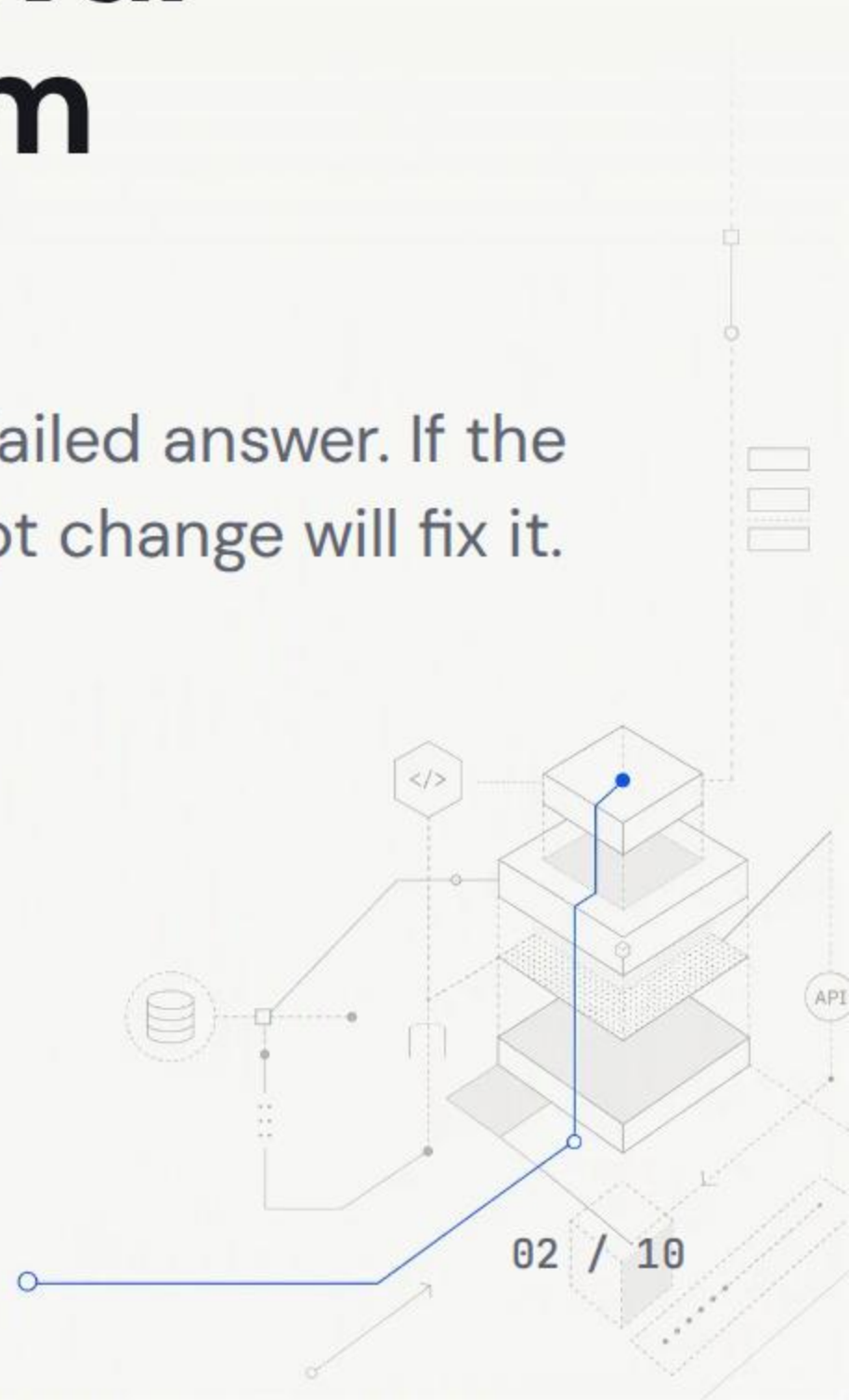
M. Ahsan Izhar - [ahsanizhar.com](https://ahsanizhar.com)



# 1

## Measure retrieval separately from generation.

Log the top-k chunks for every failed answer. If the right chunk is missing, no prompt change will fix it.



# Embeddings

## CONCEPT

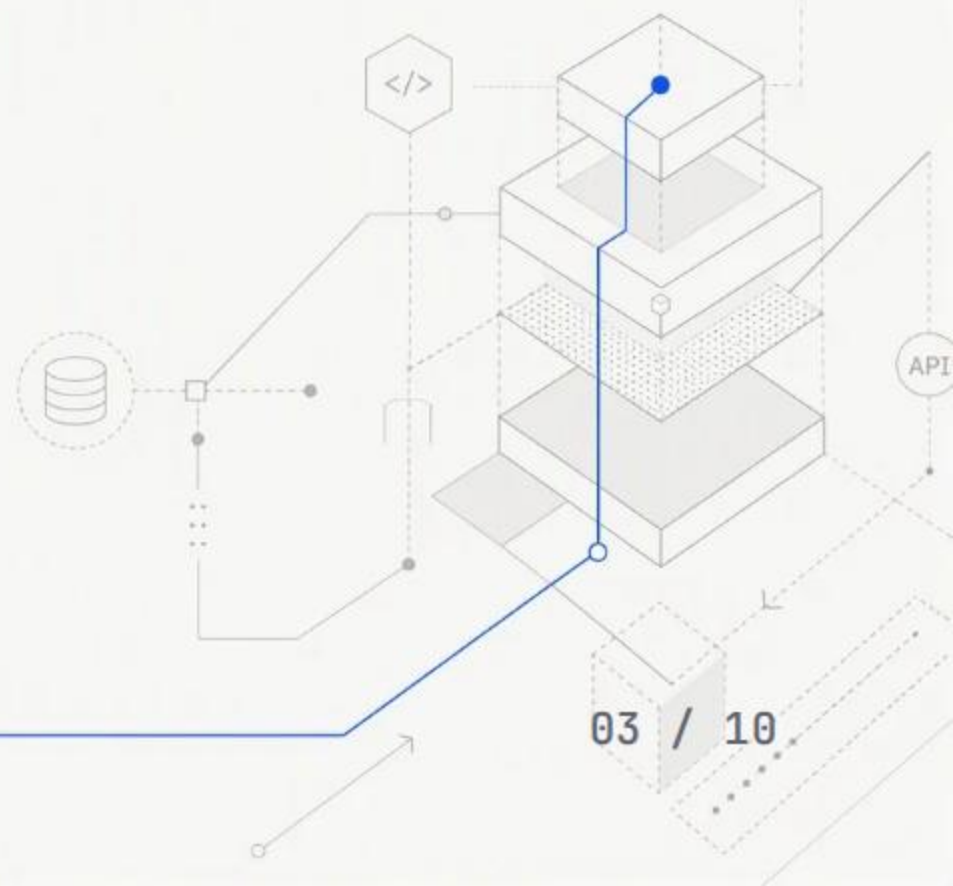
Text turned into a vector, so similar meaning lands close together.

## MECHANISM

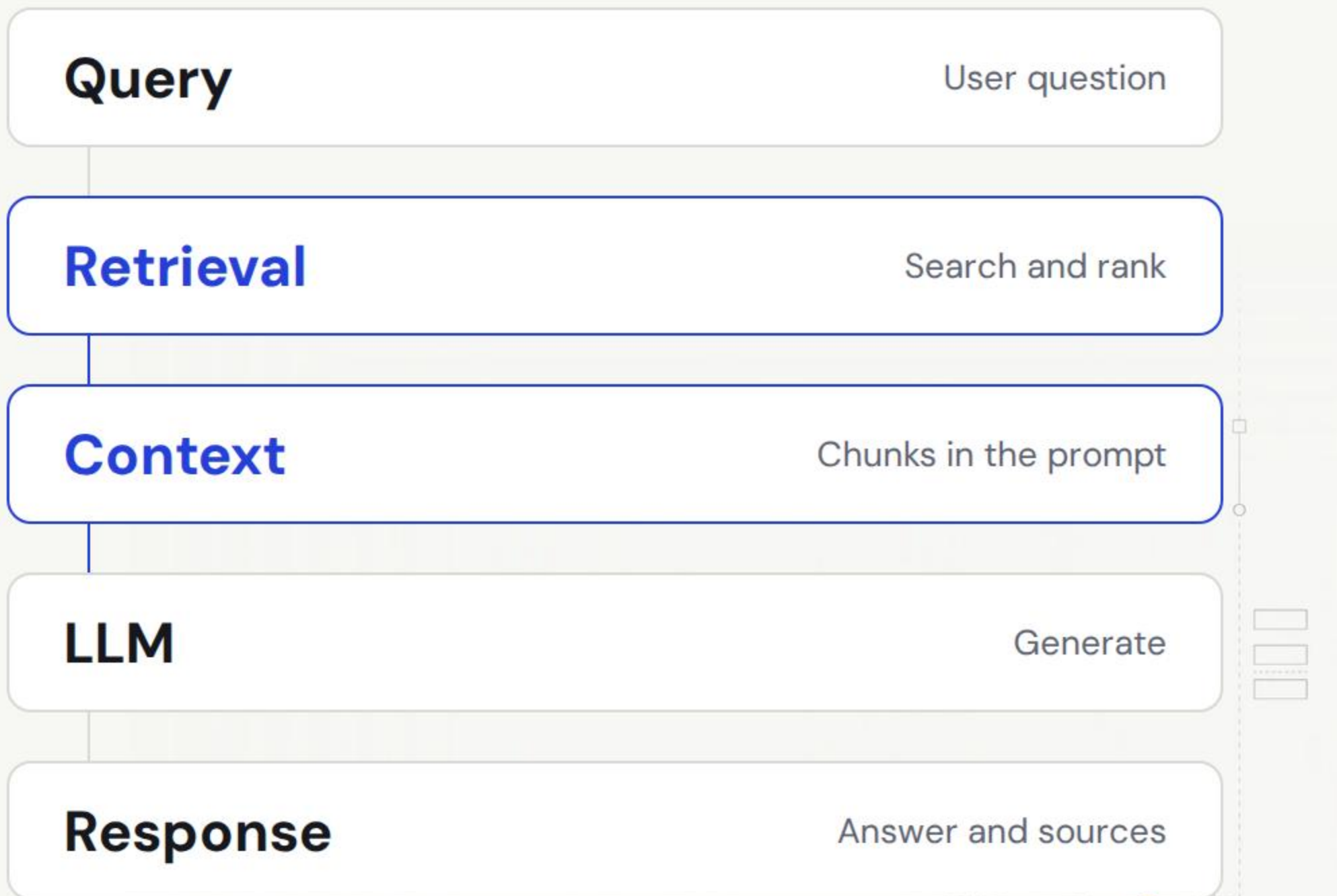
A model maps each chunk to numbers. Search ranks chunks by distance to the query vector.

## TAKEAWAY

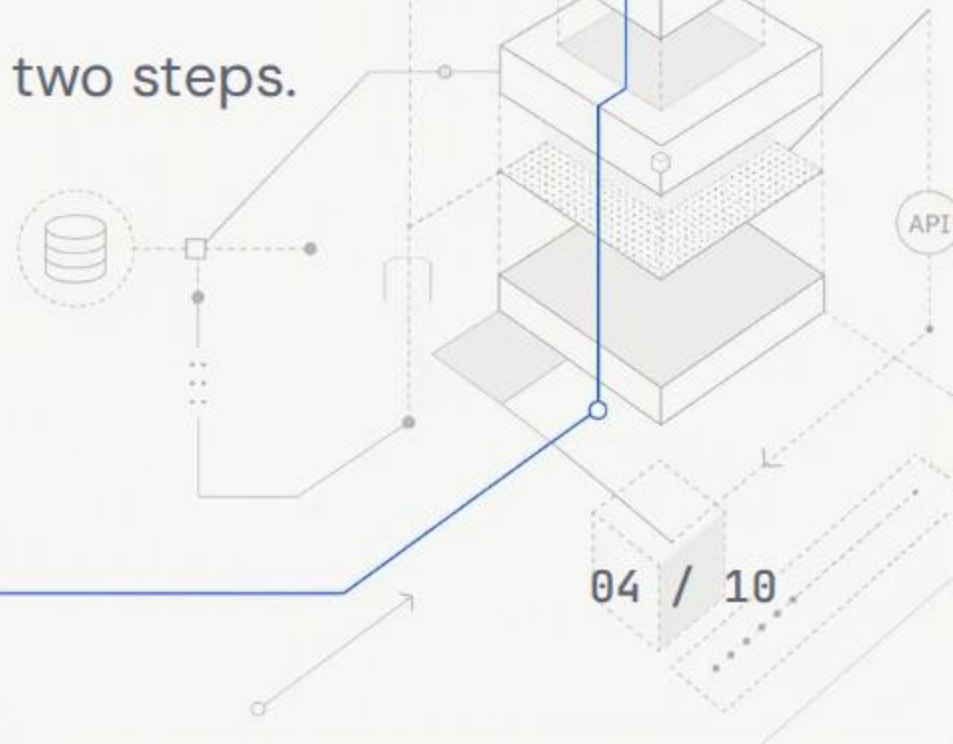
**Similar is not the same as relevant.**  
**Test with real queries.**



# How a RAG request flows



Blue path: most failures start in these two steps.



# RAG vs fine-tuning

## RAG

Knowledge changes often

You need citations

Data stays outside the model

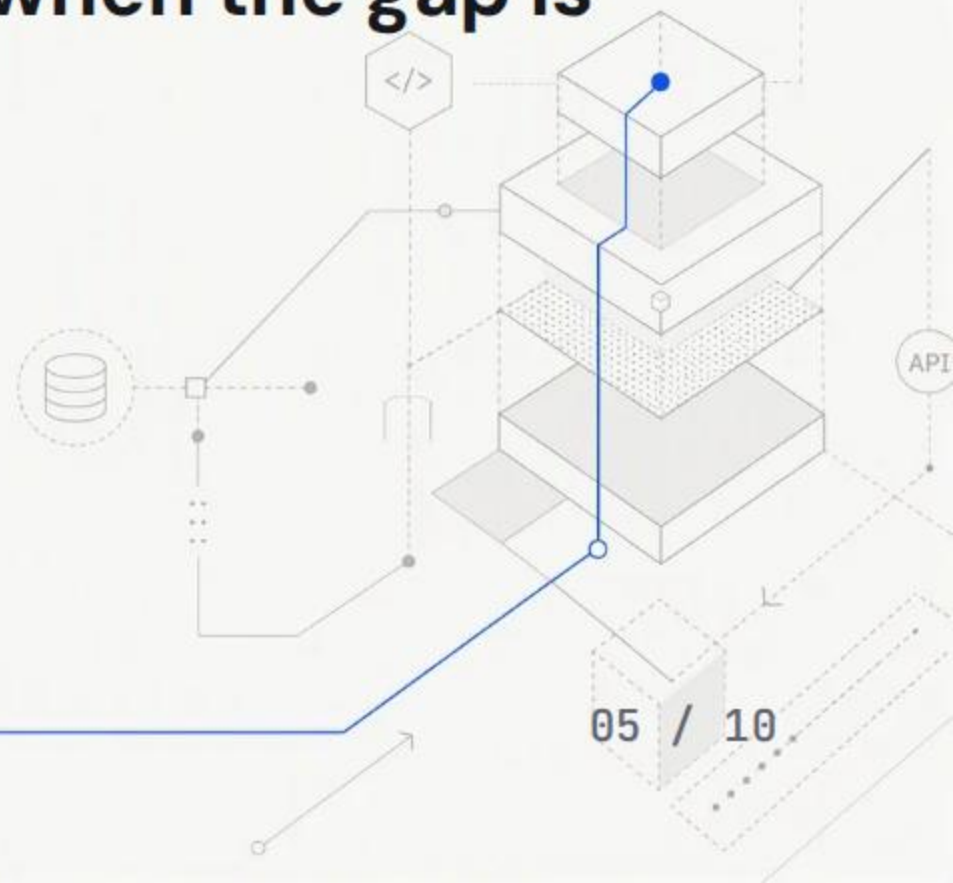
## Fine-tuning

Output format must be consistent

Per-call cost and latency matter

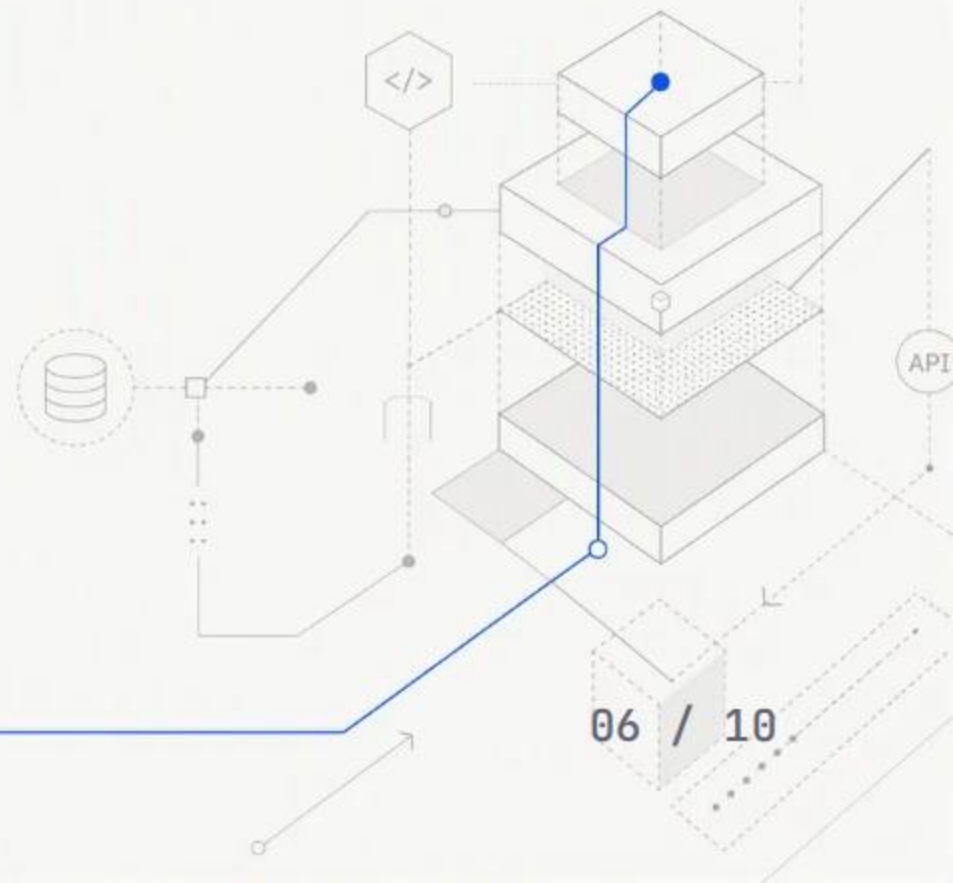
Knowledge is stable

**Start with RAG. Fine-tune when the gap is behavior, not knowledge.**



# An agent is a loop with tools. Reliability lives in how it fails.

Timeouts, malformed tool arguments, and retries deserve design time.



# 412<sub>ms</sub>

p95 retrieval latency before caching

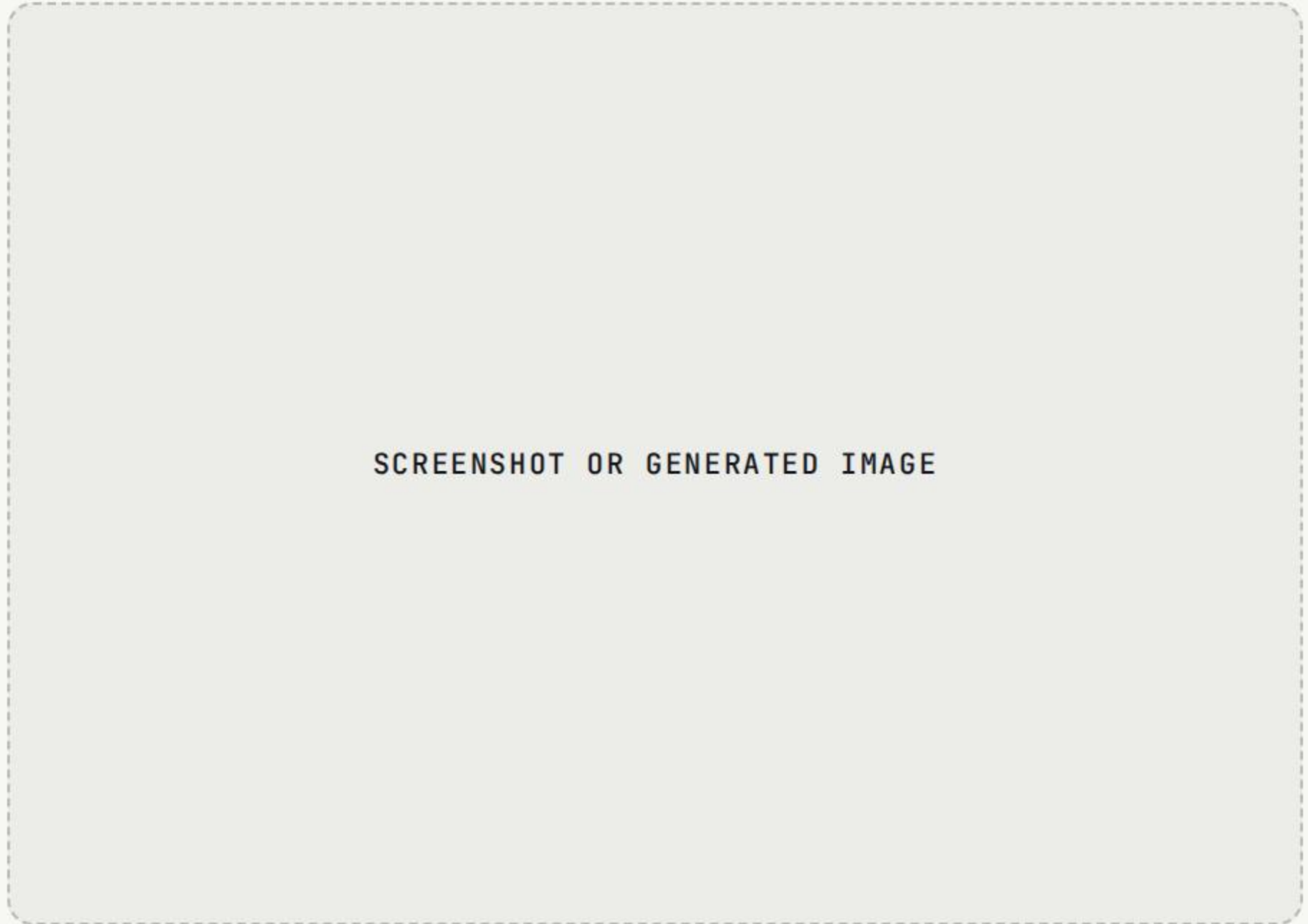
## RECALL@5 BY STRATEGY



# One tool call, fully specified

```
{
  "name": "search_docs",
  "arguments": {
    "query": "refund policy",
    "top_k": 5
  }
}
```

**Validate before you execute.** The model's JSON is input, not a trusted contract.



SCREENSHOT OR GENERATED IMAGE

FIG. 1 RETRIEVAL TRACE, FAILED QUERY

## Read the trace first.

The wrong chunk ranked first. The prompt was fine.

# Fix retrieval first. Then tune the prompt.

What would you add?

Save this for later.

---

**M. Ahsan Izhar**

Software Engineering × AI Integration

M. Ahsan Izhar - ahsanizhar.com

